Integrated Entrainer Design for Extractive Distillation using The Infinitely Sharp Splits and Computer-Aided Molecular Design

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Abstract. Entrainers play an important role in ensuring thermodynamic feasibility and economic viability of extractive distillation. Conventional selection methods, based on heuristics, focus on a limited number of commonly used entrainers, with a risk of suboptimal entrainer selection. Here, we provide an integrated framework for entrainer screening that combines: A computer-aided molecular design (CAMD) procedure used to screen entrainers based on the pure compound/mixture properties and a short-cut method based on the infinitely sharp splits (ISS) concept to evaluate key process parameters such as the minimum entrainer-to-distillate ratio and the corresponding minimum reflux ratio. The framework is successfully applied to screen and evaluate commonly used entrainers for the azeotropic mixture of acetone-methanol with results in agreement with optimization studies, where DMSO showed overall best performance w.r.t energy consumption and total annual costs compared to entrainers such as water and ethylene glycol.

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Introduction

Extractive distillation is used to separate non-ideal mixtures with azeotropes present [1,2]. The process requires the addition of an entrainer capable of interacting preferentially with the azeotropic mixture members (A-B), thereby altering the relative volatility [2]. A classical process flowsheet for extractive distillation consists of an extractive column and a recovery column, as shown in **Fig. 1**.

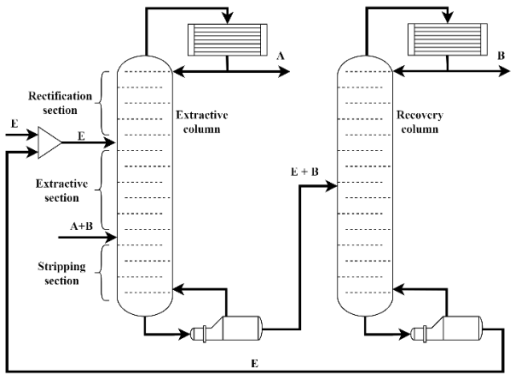


Fig. 1 Typical flowsheet for extractive distillation process with heavy entrainer

Here, the entrainer is fed into the extractive column at a tray above that of the azeotropic feed mixture (A-B) to maximize the exposure of the mixture to the entrainer. By altering the relative volatility, component A can be recovered as the top product of the extractive column, and B (also referred to as the solute) is recovered as the top product of the recovery column (**Fig. 1**, right), thus allowing separation of the azeotropic mixture. However, such benefits are also accompanied by an increase in the degree of complexity by adding another column, plus various degrees of freedom and design choices that ultimately influence process feasibility and operability [2]. A challenging but important task

is the identification of a suitable entrainer, as it plays a vital role in ensuring the thermodynamic feasibility and economic viability of the downstream process [1].

We resolve entrainer selection methods into heuristic approaches and computer-aided molecular design (CAMD) approaches [2]. Relying solely on the measurement of equilibria is not so widespread today. Measurements are frequently difficult and costly and are therefore kept to the minimum. It is more common to employ heuristics toward some reasonable processing scheme and for data to serve as a basis for screening estimates. It is infeasible to obtain phase data for all combinations of components, interpolations and extrapolations of data are routinely made instead. Heuristic approaches are based on evaluating a set of thermodynamic considerations [3,4] involving selectivity and capacity. The purpose of the entrainer is to displace the relative volatility away from unity. For a fixed temperature and pressure, the influence of the entrainer on the relative volatility can be expressed in terms of the ratio of the activity coefficients i.e. the selectivity. The capacity is an indication of the strength of the affinity of solute for the entrainer [2]. Pierotti and coworkers developed extensive systematic studies of infinite dilution activity coefficients of homologous series [3]. They considered the infinitely dilute solution for a systematic study of group effects. There are several reasons for emphasizing the dilute region, in particular, non-ideality effects are usually at their maximum in the dilute regions. Thus, the maximum solution effects are generally observed directly by the limiting values at infinite dilution. In other words, the selectivity at infinite dilution can be expected to be indicative of a maximum entrainer effect on relative volatility. Further, the infinitely dilute state is a well-defined common point of reference, making it suitable for comparisons. Extensive experience exists with thermodynamically consistent correlations to provide interpolation into regions of finite concentrations. Thus, we evaluate selectivity and capacity at infinite dilution, as shown in **Eq 0.e (Fig.2**) and **Eq 0.f (Fig.2**). It is desirable for the separation in the first column. that selectivity is large. That will be the case if the capacity of the entrainer for A is low and the capacity for B is high. Other heuristic measures can employ molecular classifications by evaluating the ability of the entrainer to form hydrogen bonds with the solute [2]. Concepts obtained from heuristic approaches have been incorporated into a systematic and computer-based approach in the form of CAMD [2,5]. In CAMD, a set of molecular building blocks in the form of functional groups are selected and assembled systematically to generate molecules satisfying a set of structural and property constraints and evaluated in terms of an objective function [6]. CAMD can be formulated as a mixed-integer Nonlinear problem (MINLP) [2,5,6]. The property constraints reflect the set of demands placed on the entrainer. These can either be pure compound properties or mixture properties. These properties are usually evaluated using group-contribution-based property models [6]. Most approaches consider objective functions, based on the molecular properties of the system such as selectivity, capacity, or both, while the actual process performance is not considered. Other concepts can be incorporated in evaluating the process performance. For example, a high driving force of separation would correspond to an easier separation, particularly for the recovery column [5], making it a candidate for consideration. Commonly the process evaluation is done as a post-processing step, using process simulation. This can be a challenging optimization problem [2]. In this work, we propose a CAMD approach integrated with a short-cut method that provides process insight by evaluating two key process parameters, namely the minimum entrainer-to-distillate ratio and the corresponding minimum reflux ratio. The scope is that compared to process simulation, the short-cut method can provide process insights using calculations that are both fairly simple and rapidly applied.

Integrated Framework for Computer-Aided Molecular Design and Process Evaluation

The equations can be seen in **Fig. *2***. The choice of molecular building blocks is crucial to ensure that the pure and mixture properties of the system can be evaluated. Widely used functional groups that allow evaluation of all properties, constraints, and criteria are the UNIFAC [7], and the Constantinou-Gani [8] groups. Feasible group sets must satisfy a set of *structural connectivity constraints* shown in (**Eq. 1.a**) and (**Eq. 1.b**), where v_i is the valency and n_i is the occurrence of the group "i" [9]. Additional structural constraints include a limit on the number of times a specific group is allowed to occur (**Eq. 1.c**) and the total number of groups in a molecule (**Eq. 1.d**). In addition, the entrainer should satisfy the *pure component property constraints* such as being liquid (**Eq. 2.a**) and being the compound with the highest boiling point (**Eq. 2.b**). Additional constraints can involve a constraint on the heat of vaporization (ΔH_v, **Eq. 2.c**). The pure component properties are calculated using the Constantinou-Gani group contribution methods (**Eq. 0.a**, **Eq. 0.b, and Eq. 0.c**), where n_i is the occurrence vector of the group "i", T_{mi}, T_{bi}, and ΔH_{vi} are the contributions of the group "i" and T_{m0}, T_{b0} and ΔH_{v0} are empirical model constants [8]. The entrainer should, furthermore, satisfy a set of *mixture property constraints* on the phase behavior of the system. High while selectivity is desirable, it is not part of the objective function but rather used to screen (cut-off criterion) (**Eq. 3.a**). The entrainer should also be fully miscible with components A and B. This is ensured by restrictions on the Gibbs free

energy of mixing and its second derivative (**Eq. 3.b** and **Eq. 3.c**) [10]. Furthermore, the entrainer should not form any new azeotropes with either component A or B. This is determined by evaluating the relative volatility at each end of the binary VLE compositions (**Eq. 3.d** and **Eq. 3.e**). The vapor pressure is determined by a two-parameter correlation fitted to the conditions at the boiling and critical points. This is less accurate than could be wished for, but it allows fully predictive screening and the level of approximation is similar to all components considered. The novelty of the developed framework is the integration into CAMD of a short-cut method based on the concept of infinitely sharp split (ISS) occurring in the extractive column [11]. The method is based on pinch analysis to determine the feasibility of extractive distillation at finite reflux ratios and an infinite number of trays [1]. The short-cut model is considered an optimization problem to determine a composition (x^{r}_{A}) along the A-E branch of the ternary diagram for which the entrainer-to-distillate (E/D) is minimum **(Eq. 4.a** and **Eq. 4.b)**. This composition must be below the composition corresponding to the intersection of the univolatility curve ($\alpha_{A,B}$=1) at the binary edge A-E (x^{iso}_{A}) (**Eq. 4.d** and **Eq. 4.e**). The corresponding reflux ratio (R_{min}) can then be determined using **(Eq. 4.c)**. The details of the method are covered at greater length in a companion paper of this conference [12]. When performing the VLE calculations, the gas phase is assumed to be an ideal gas mixture and UNIFAC is used to model the liquid activity coefficient [7]. The CAMD problem can be considered an optimization problem that is solved using the "generate-and-test principle" and by decomposing the constraints into blocks of constraints and evaluating them sequentially [13]. This allows for full characterization and exploration of the design space defined. The addition of the ISS method produces a nested optimization problem to identify the composition resulting in $(E/D)_{min}$. The final molecules can then be ranked either based on the $(E/D)_{min}$ or R_{min} or as a combination of both by considering the CAMD as a multi-objective optimization problem. Industrial design usually set first the operating (E/D) from its minimum value and then the extractive distillation column is designed (total trays, feeding positions and reflux ratio) to minimize total cost. Therefore, we perform the ranking using $(E/D)_{min}$.

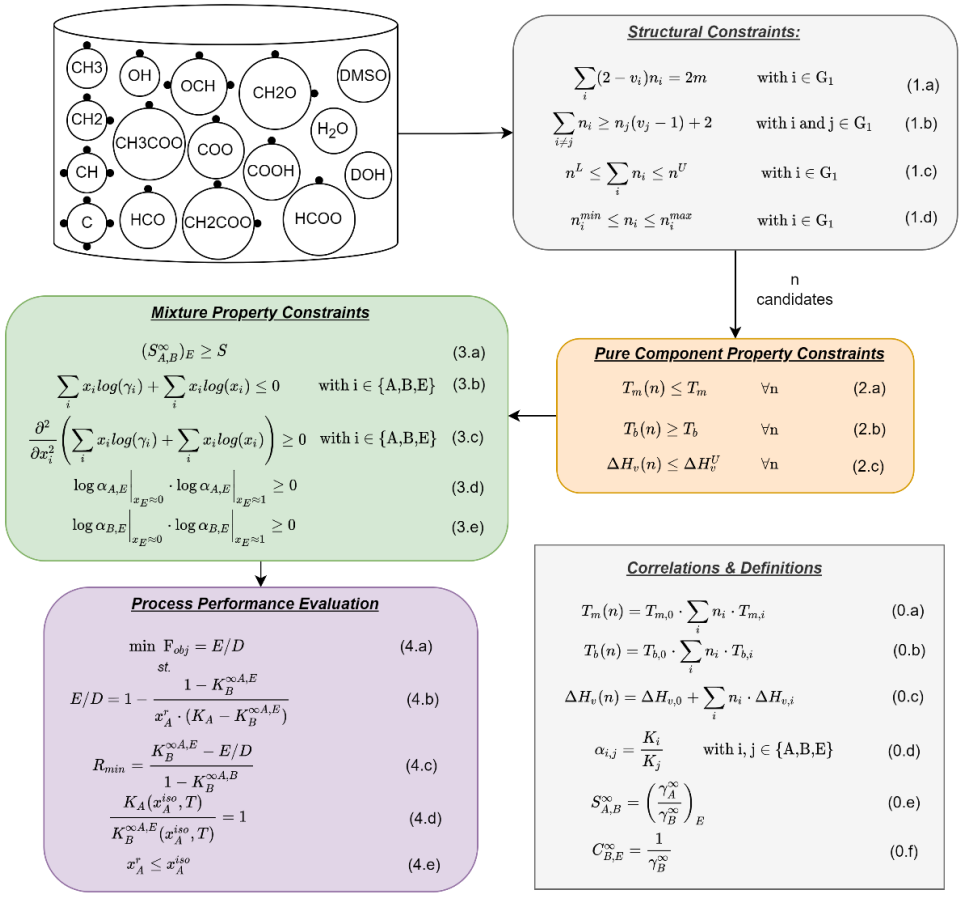


Fig. 2 Integrated Framework for the design and evaluation of heavy entrainer for extractive distillation purposes

Case Study: Separation of acetone-methanol using heavy entrainers

The binary system acetone-methanol is an established system in terms of the design and optimization of extractive distillation processes with various established benchmark entrainers [2,5]. Both compounds are the product of the Fischer-Tropsch process using carbon monoxide and hydrogen to produce hydrocarbons [1]. The mixture exhibits a minimum boiling azeotrope near 328 K with an acetone composition near 0.79 at 1 atm. Depending on the entrainer selected, either acetone or methanol can be the top product of the extractive column. In this study, we focus on heavy entrainers for which acetone is the top product of the extractive column (A: acetone, B: methanol). Two studies are presented:

1) **Study 1: Evaluating existing benchmark entrainers**
2) **Study 2: Generating and evaluating new entrainer candidates through molecular design**

For both studies, the upper and lower bonds are given in **Table 1**. Important to note is that the structural connectivity constraints are not active when evaluating benchmark entrainers.

Table 1. Lower and upper bounds for various needs and constraints

Quantity	Constraint	Lower	Upper	Quantity	Constraint	Lower	Upper
Nr. of groups	(1.c)	3	6	T_b(K)	(2.b)	368	500
Nr. of same groups	(1.d)	1	2	ΔH_v(kJ/mol)	(2.c)	-	67
T_m(K)	(2.a)	-	298	S	(3.a)	2	-

Study 1: Evaluation of existing benchmark entrainers

Water, DMSO, and ethylene glycol (DOH) were selected as benchmark entrainers. The continuous reduction of the chemical design space resulting from applying the framework can be seen in **Fig. 3**. All entrainers satisfy the pure property constraints. However, DMSO and water do not satisfy the mixture constraints. This is due to strong demands being placed on the selectivity in the case of DMSO. In the case of water, UNIFAC predicts the formation of a (false) azeotrope between water and acetone at very high acetone compositions. Nevertheless, validation studies exist demonstrating that water and DMSO are promising entrainers [2,5]. This suggests that these constraints are very sensitive to the accuracy of thermodynamic properties, and relaxation is needed to avoid eliminating promising candidates. We, therefore, conclude that it is better to include processing constraints that emphasize the behavior near realistic processing conditions. Such information must incorporate process models, and yet be computationally inexpensive. Concepts based on short-cut models could potentially provide a promising alternative.

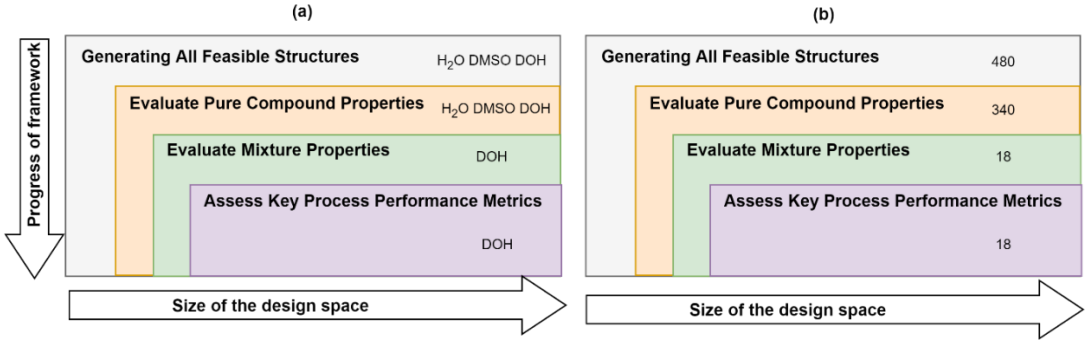


Fig. 3 Continuous reduction of the chemical design space: a) for benchmark entrainers b) for the CAMD procedure

The selectivity at infinite dilution, the capacity, the minimum entrainer to distillate ratio, and the corresponding reflux ratio for the benchmark entrainers are shown in **Table 2**.

Table 2. Selectivity, capacity, minimum entrainer to distillate ratio, and corresponding reflux ratio for benchmark entrainers

Rank	Entrainer	S^{∞}	C^{∞}	$(E/D)_{min}$	R_{min}
1	Water	5.24	0.94	0.020	3.62
2	DMSO	1.46	0.80	0.619	1.38
3	Ethylene Glycol	2.24	0.71	1.021	1.38

Table 2 shows that water exhibits the highest selectivity. Yet, the potential energy requirement resulting from the high reflux ratio makes it a less desirable option. Meanwhile, DMSO exhibits a similar reflux ratio compared to DOH but with a much lower entrainer-to-distillate ratio. Both factors will translate into a lower total annual cost (TAC) which was confirmed through a rigorous design and economic evaluation in a previous study [2,5]. These results further consolidate previous findings that exclusively using the selectivity at infinite dilution as the selection criteria might erroneously eliminate promising candidates and thus a proper short-cut model to assess the performance of the entrainer is needed [2].

Study 2: Generating & evaluating new entrainer candidates using molecular design

The framework is switched to "generate-and-test mode". For the current case study, only acyclic solvents (m=1 in (**Eq. 1.a**)) exclusively made up of C, H, and/or O are considered. The groups considered are thus {CH_3, CH_2, CH, C, OH, CH_3COO, HCO, OCH, CH_2O, COO, CH_2COO, COOH, HCOO}. The user can easily modify the groups considered to only include chemical species of special interest. The specific limits on the constraints are summarized in **Table 1**. A total of 480 structurally compliant compounds were generated, of which 340 candidates satisfied the pure compound constraints. However, only 18 compounds were compliant with the mixture property constraints and are thus considered for the ISS short-cut method evaluation. The progressive reduction of the design space is illustrated in **Fig. 3**. The groups, potential identity of the compound as well as selectivity, capacity, entrainer-to-distillate ratio, and the corresponding entrainer candidates can be seen in **Table 3**. Based on these results, many entrainer candidates could potentially outperform DMSO and the remaining benchmark entrainers when considering the $(E/D)_{min}$ and R_{min}. However, the CAMD procedure does not take into account aspects such as how easy it is to synthesize the compound, the cost of the entrainer nor how easy the separation is in the recovery column. Similarly, the stability of molecules satisfying connectivity constraints is not guaranteed. **Table 3** has some examples of molecules that are questionable in this respect such as 2-hydroxypropanal and propandial. This could be addressed by including constraints on the connectivity rules such as the number of times a single group can be selected in a feasible structure. However, we have not delved too deeply into these aspects here.

Table 3. Selectivity, capacity, minimum entrainer to distillate ratio, and corresponding reflux ratio for top entrainer candidates from the molecular design framework ranked based on $(E/D)_{min}$

Rank	Groups						Compound Name	S^{∞}	C^{∞}	$(E/D)_{min}$	R_{min}
	CH_3	CH_2	CH	C	OH	HCO					
1	0	1	0	0	0	2	Propanedial	3.97	2.58	0.26	1.15
2	1	0	1	0	0	2	2-methylpropanedial	2.75	2.56	0.42	1.57
3	1	0	1	0	1	1	2-Hydroxypropanal	2.94	1.91	0.42	1.57
4	0	2	0	0	0	2	Butanedial	2.68	2.50	0.48	1.66
5	1	0	1	0	2	0	1,1-Ethanediol	2.78	1.18	0.50	1.46
6	2	0	0	1	0	2	2,2-dimethylpropanedial	2.18	2.41	0.53	1.95
7	0	1	0	0	1	1	2-hydroxyacetaldehyde	4.01	1.79	0.57	1.18
8	1	1	1	0	0	2	2-methyl-1,4-butanedial Ethylmalonaldehyde	2.16	2.37	0.58	2.00
9	0	2	0	0	1	1	3-Hydroxypropanal	2.87	1.89	0.59	1.54
10	0	3	0	0	0	2	Pentanedial	2.12	2.33	0.61	2.06
12	**0**	**2**	**0**	**0**	**2**	**0**	**Ethylene glycol**	**2.74**	**1.18**	**0.70**	**1.53**

Entrainer candidate ranked (12) is included in **Table 3** corresponds to ethylene glycol generated using the CAMD procedure. The UNIFAC model considers ethylene glycol as a single group molecule rather than the combination of

CH_2 and OH groups largely due to substantial proximity effects present [7]. Although the selectivity, capacity, and reflux ratio are comparable between **Table 2** and **Table 3**, the difference in E/D is notable. This indicates that proximity effects could potentially alter the results and potential conclusions. However, considering that the tool is intended for screening purposes such discrepancies can be accepted and dealt with as part of the post-processing of the results.

Conclusion

We provided an integrated product-process design framework for screening entrainers for extractive distillation based on a molecular design using group contributions and the ISS short-cut method for process evaluation. The ISS short-cut method is capable of determining key process parameters to provide insights into the economics of the process. The tool was tested on the case study of identifying entrainers for the separation of an acetone-methanol mixture showing results consistent with rigorous process optimization studies. The framework provided is quite flexible and can be modified to evaluate entrainers for other azeotropic mixtures as well as including other considerations such as safety and flammability-related properties such as auto-ignition and flashpoint temperatures [13]. However, the general purpose of the framework is to provide a tool for screening that allows practitioners to focus on a subset of the chemical design space and potentially discover better alternatives. This was demonstrated through the case study of identifying entrainers for the separation of an acetone-methanol mixture, and it is a key trade-off in developing decision tools for the design of such processes, especially when using group-contribution based methods. The tools, however, do not eliminate the necessity of a post-processing step where the groups chosen to form a compound are inspected for their stability, availability, and synthesizability. Finally, rigorous simulations are a natural step after the application of the tool to validate the results obtained. For the second case study, this is left for future work as the main highlight is the integration of the ISS method in a CAMD framework.

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